BLOCKCHAIN Experiment No: 8

Harshvardhan Patil
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Aim

To configure a cryptocurrency wallet using MetaMask and integrate it with a decentralized application (DApp) to perform and verify blockchain transactions.

Theory

MetaMask is a cryptocurrency wallet that provides users with an interface to interact with blockchain networks and decentralized applications. It manages blockchain accounts and allows users to approve or reject transactions initiated by DApps.

A Decentralized Application (DApp) is an application that interacts with blockchain-based smart contracts. The frontend of a DApp communicates with the blockchain through a Web3 provider such as MetaMask. MetaMask acts as a bridge between the user's browser and the blockchain network.

In this experiment, the ZeroTrace DApp was used to demonstrate MetaMask integration. The DApp uses Web3 technologies to connect the user's wallet and interact with deployed smart contracts. After connecting MetaMask, a blockchain transaction was initiated through the DApp. MetaMask displayed the transaction request, which was confirmed by the user. The resulting transaction was then verified using its transaction ID on a blockchain explorer.

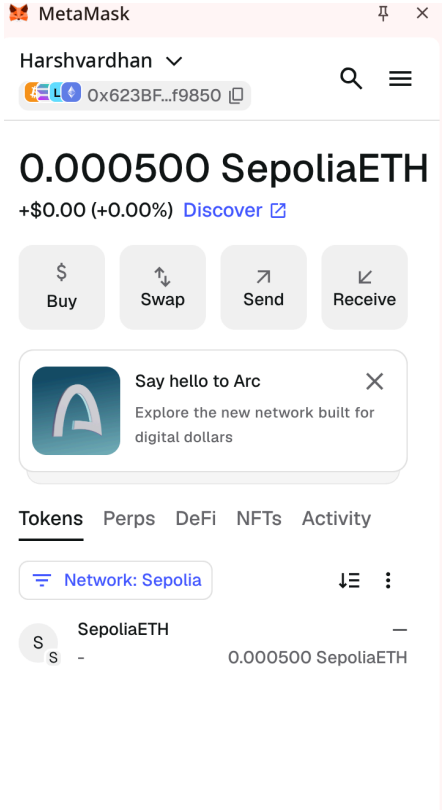
The basic interaction can be represented as:

User
↓
ZeroTrace DApp
↓
MetaMask Wallet
↓
Smart Contract
↓
Blockchain Network
↓
Transaction Confirmation

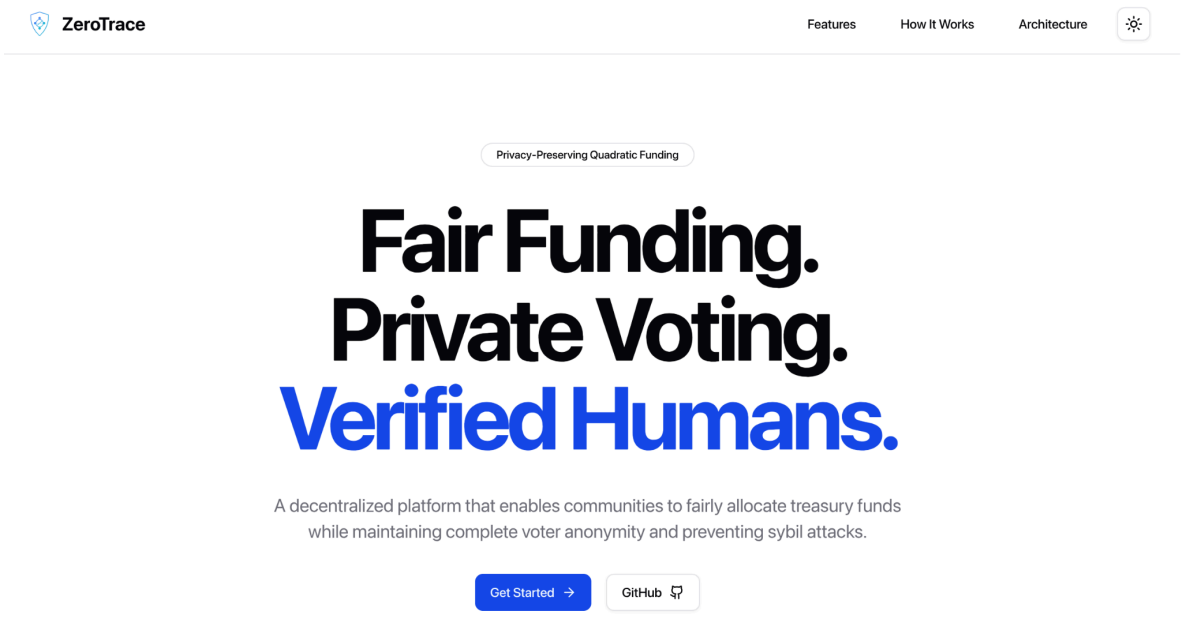
MetaMask provides important functions such as account management, network selection, transaction signing, and connection authorization. A transaction hash is generated for a submitted blockchain transaction and can be used to verify its status and details on a blockchain explorer.

Output / Screenshot Description

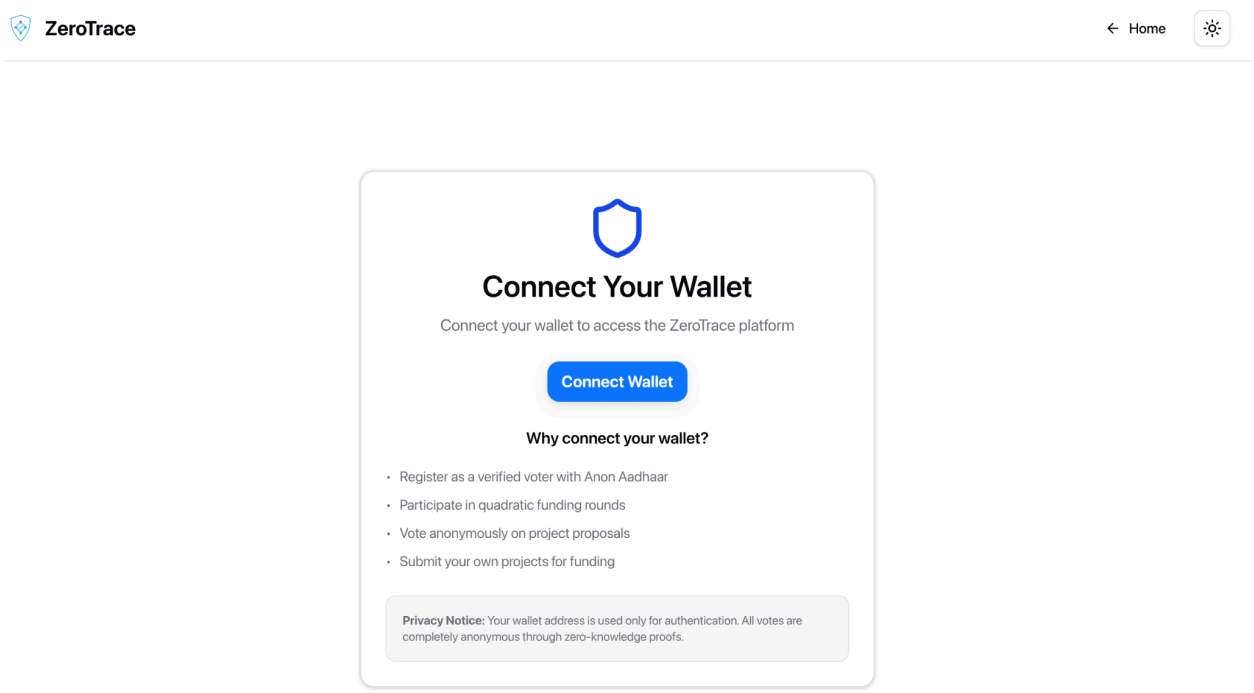
Screenshot 1: MetaMask Wallet Account



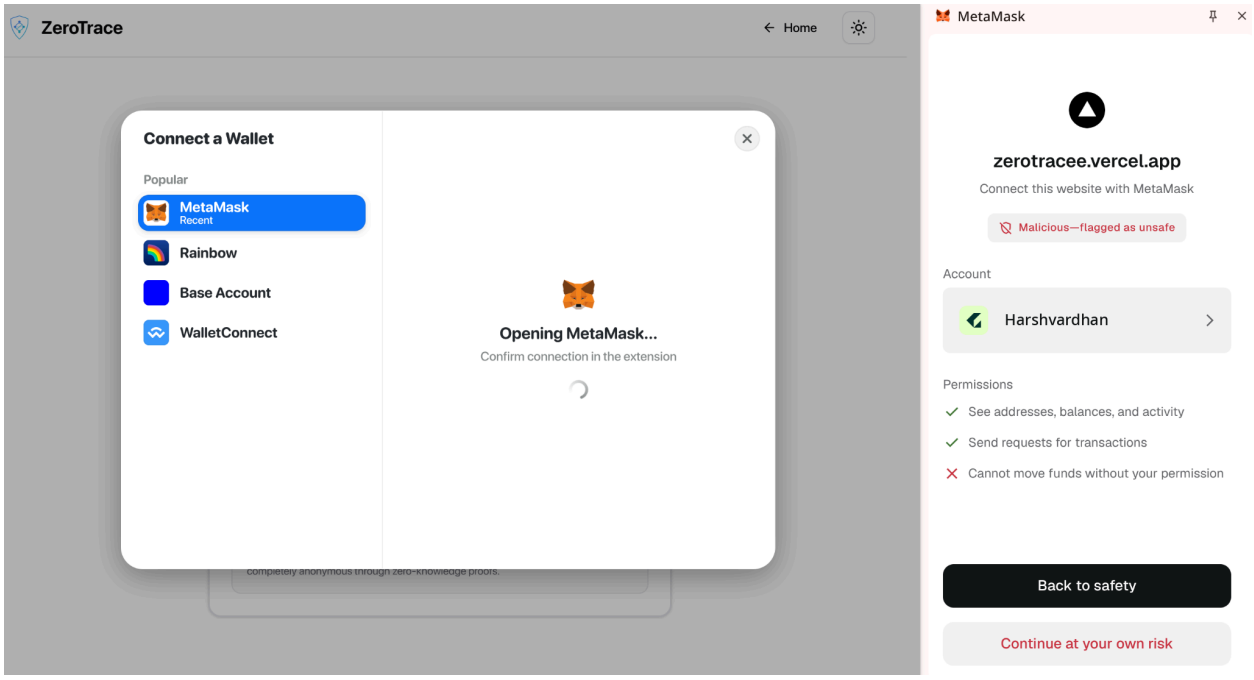
Screenshot 2: ZeroTrace DApp



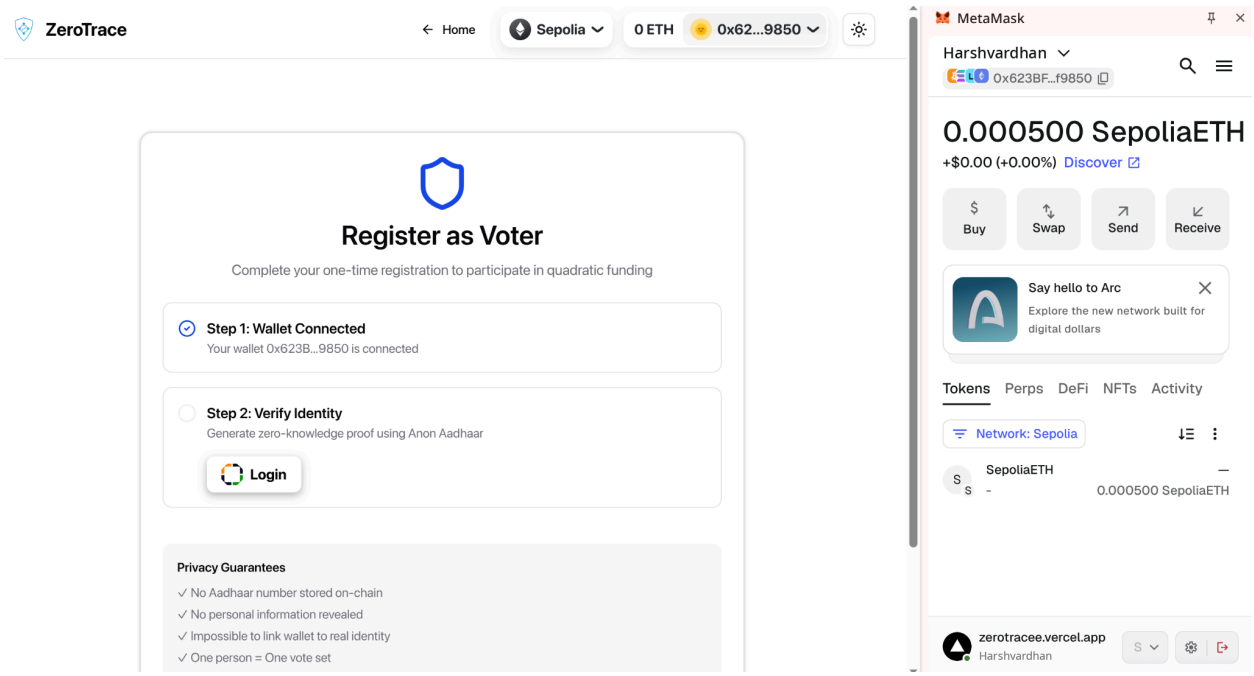
Screenshot 3: Connect Wallet Option



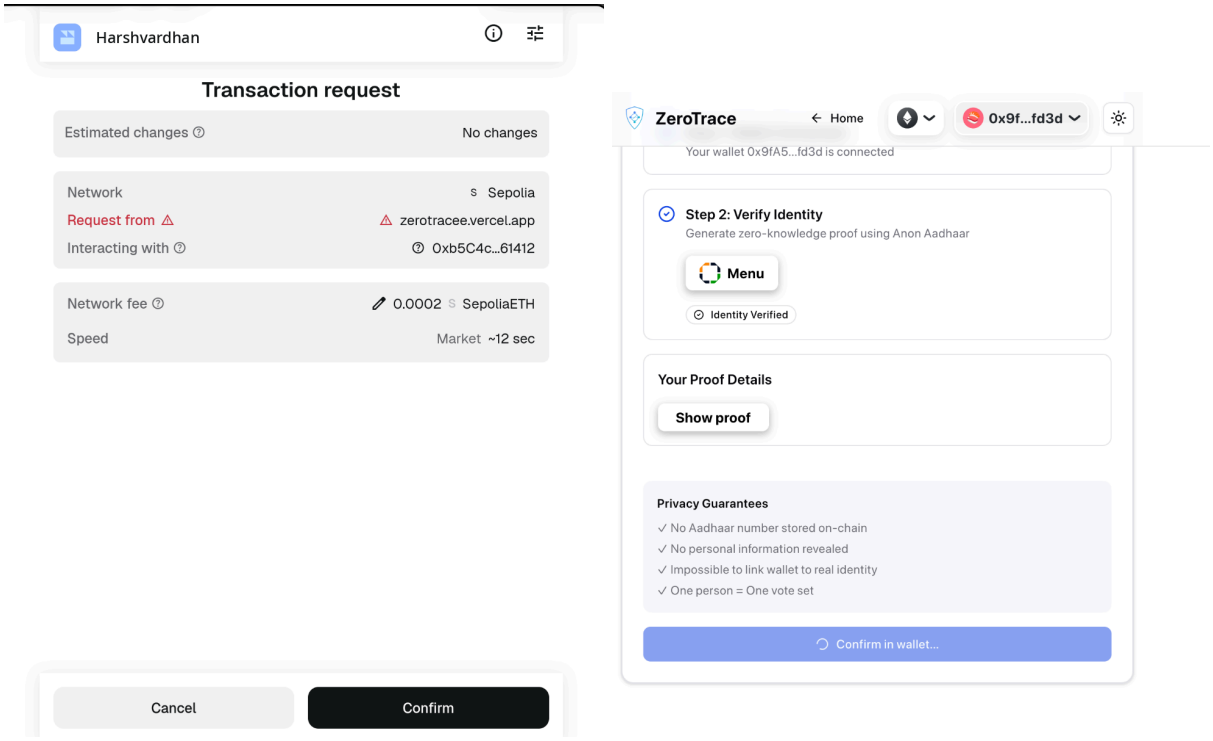
Screenshot 4: MetaMask Connection Approval

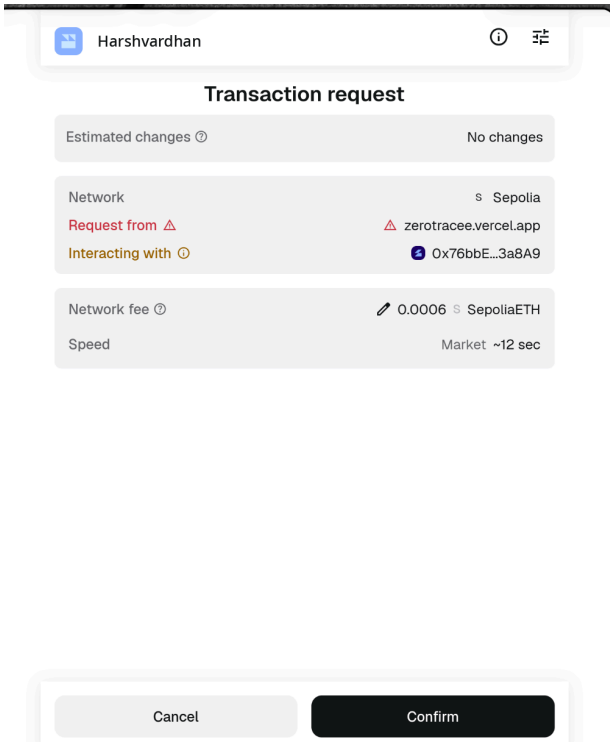


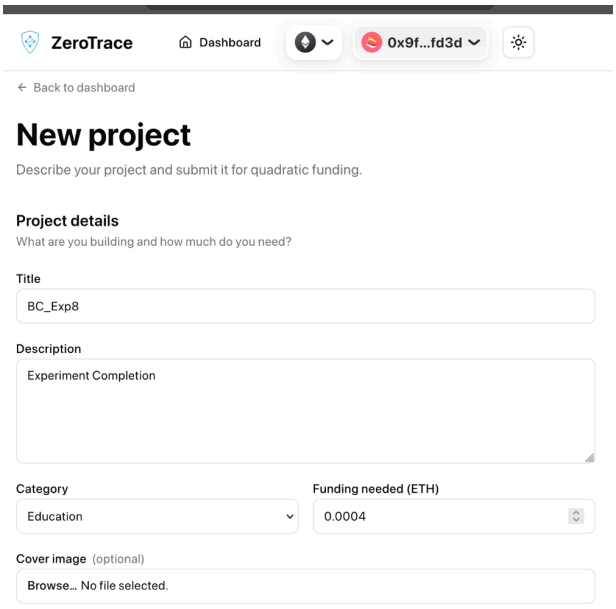
Screenshot 5: Successfully Connected Wallet



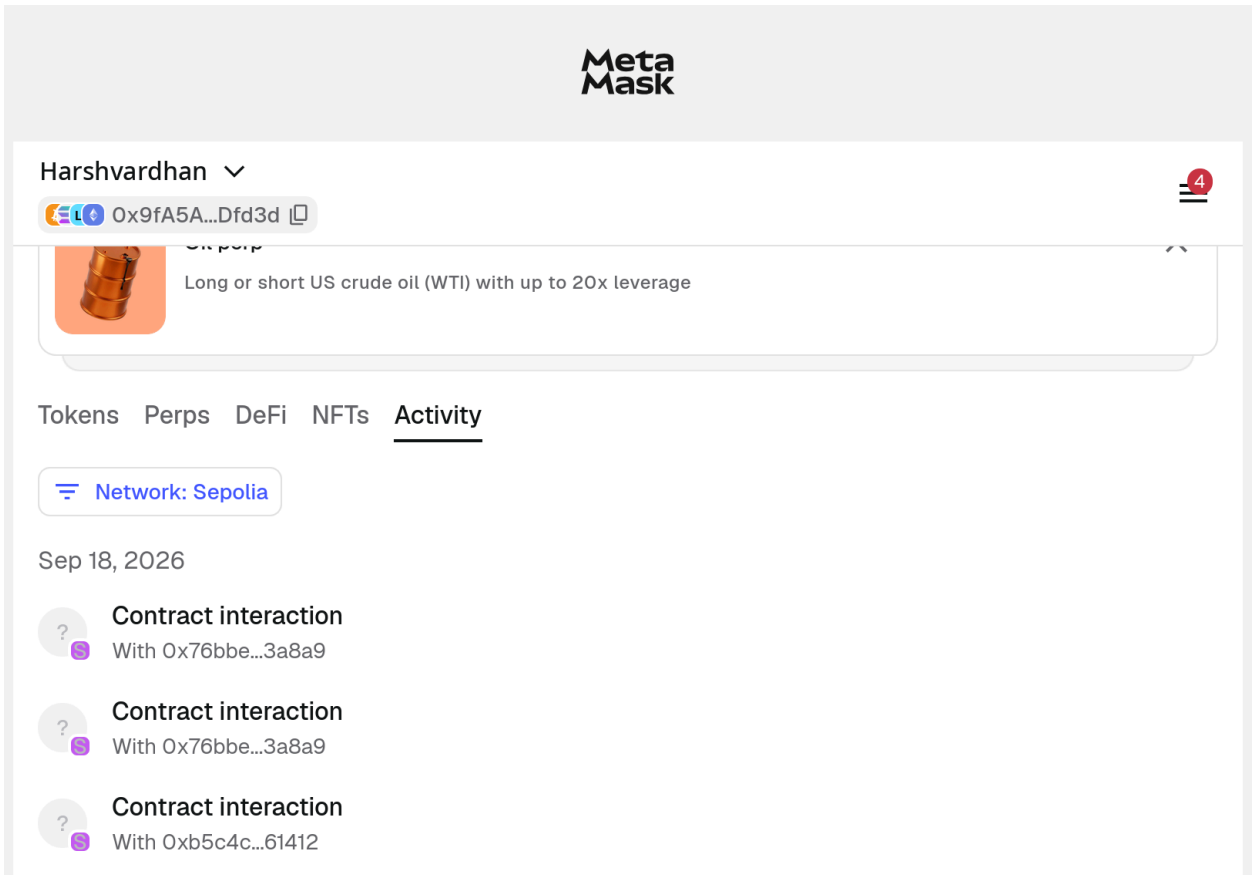
Screenshot 6: MetaMask Transaction Confirmation



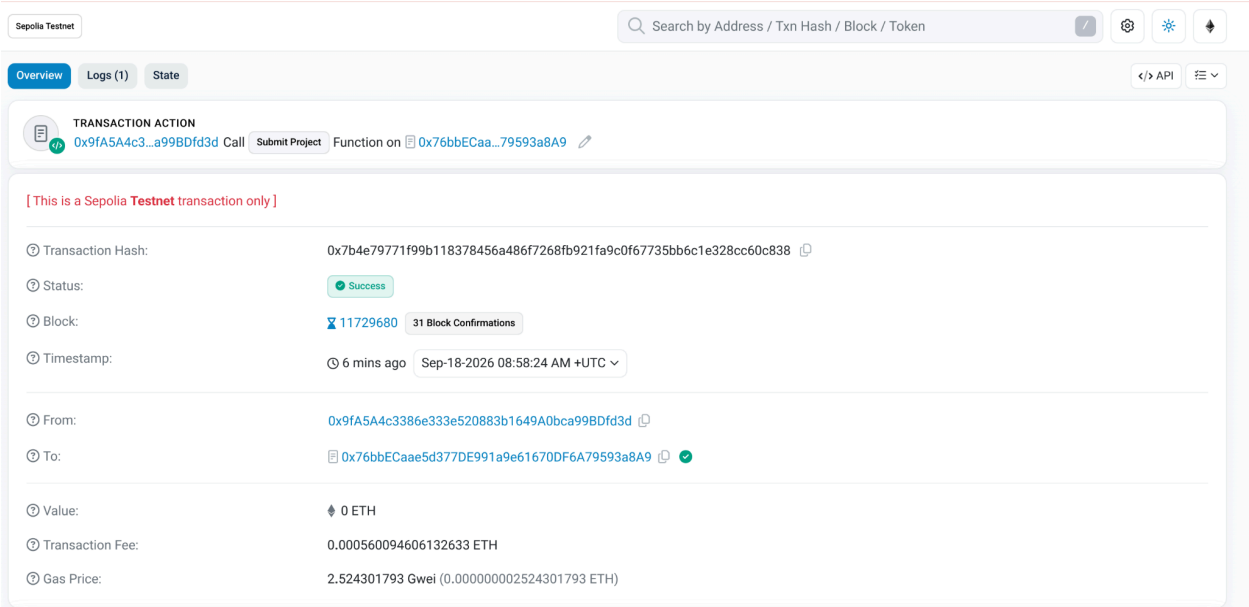




Screenshot 7: Successful DApp Transaction



Screenshot 8: Transaction Verification



Conclusion

Thus, the MetaMask cryptocurrency wallet was successfully configured and integrated with the ZeroTrace DApp. The DApp was able to connect to the MetaMask wallet and initiate a blockchain transaction, which was approved by the user through MetaMask. The transaction was successfully recorded and verified using its transaction ID on the blockchain explorer. This experiment demonstrated the interaction between a cryptocurrency wallet, a decentralized application, smart contracts, and a blockchain network.